



Grade 8 Pre-Algebra

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · M/J GRADE 8 PRE-ALGEBRA · FLORIDA COURSE
1205070 · FULL YEAR

~4¼ hrs per week
50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in the lessons, examples, and practice sets from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Pre-Algebra prepares scholars for high school mathematics. The year moves from exponents, scientific notation, and the real number system into the heart of algebra: expressions, equations, functions, and linear relationships, ending the algebra arc with systems of equations. The final quarters add data and probability, then a strong geometry sequence on angle relationships, the Pythagorean Theorem, and transformations, the ideas Algebra 1 and Geometry will build on directly.

THE YEAR AT A GLANCE

M1 **Laws of Exponents & Scientific Notation** (~3 wks)
very large and very small numbers

M3 **Algebraic Expressions** (~2 wks)
monomials and equivalent expressions

M5 **Functions & Proportional Relationships** (~2 wks)
what makes a function; linear vs not

M7 **Systems of Equations** (~1½ wks)
solutions of systems, solved graphically

M9 **Probability & Simulated Experiments** (~2 wks)
predictions from repeated experiments

M11 **Triangles & Their Side Lengths** (~2 wks)
the Pythagorean Theorem

M2 **Real Numbers** (~2½ wks)
rational and irrational numbers; ordering

M4 **Equations & Inequalities in One Variable** (~2½ wks)
multi-step and radical equations

M6 **Linear Functions** (~3½ wks)
slope, graphing, slope-intercept form

M8 **Bivariate Data** (~2 wks)
scatter plots, lines of fit, predictions

M10 **Angle Relationships** (~2½ wks)
angle pairs, triangles, polygons

M12 **Transformations & Similar Triangles** (~2½ wks)
translations, rotations, dilations, similarity

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- A review and module assessment as each module closes, roughly every two to three weeks.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A scientific calculator is useful for the scientific notation and Pythagorean Theorem work.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.